



Characterization of poly(ethylene glycol)-*b*-poly(ϵ -caprolactone) by liquid chromatography under critical conditions: Influence of catalysts and reaction conditions on product composition

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ABSTRACT

Amphiphilic block copolymers were synthesised by ring opening polymerization of ϵ -caprolactone with polyethylene glycol monomethyl ether (PEG-MME) using different catalysts (boron trifluoride, sodium hydride, and tin octoate) in a one pot procedure. The products obtained were characterized with respect to their molar mass distribution and content of homopolymers using size exclusion chromatography (SEC), **Liquid chromatography under critical conditions (LCCC)** and MALDI-TOF-MS. The homopolymers of caprolactone could be separated from the block copolymer by LCCC on a reversed phase column in tetrahydrofuran–water mobile phases with evaporative light scattering detection (ELSD). Residual monomer could be determined under the same conditions using density detection, a separation of the copolymer from residual initiator could be achieved on a normal phase column in acetone–water mobile phases.

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1. Introduction

Poly(ϵ -caprolactone) (PCL) is one of the promising materials which can be applied for biomedical applications owing to its good biocompatibility, permeability and degradability [1]. Generally speaking, PCL is rather hydrophobic and it degrades very slowly by simple hydrolysis under the human body conditions. In the last years, amphiphilic block copolymers of ϵ -caprolactone (CL) and ethylene oxide (EO) have attracted increasing attention because of their special properties [2].

The presence of a polyethylene glycol (PEG) block minimizes platelet deposition onto surfaces exposed to blood

stream [3]. Surfaces having low protein adsorption have important applications, such as blood contacting implant devices, membranes for separation processes, biosensors, chromatographic supports, contact lenses, blood and protein storage applications, etc. [4].

Another important application is due to the amphiphilic character of these block copolymers. In general, amphiphilic block copolymers can self-assemble to form nano-sized micelles consisting of a hydrophilic outer shell and a hydrophobic inner core in aqueous media. Such micelles with core-shell structure can readily incorporate lipophilic drugs into their cores and release them in a controlled manner at a later stage, while the hydrophilic shell can provide stabilization for the micelles without the need for additional stabilizers, thus making them potential carriers for drugs with poor water solubility [5].

These block copolymers were employed for formation of biocompatible micelles as nanocarrier for drug delivery

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[6], encapsulation of anticancer drugs, and as thermosensitive biomaterials [7,8]. They are suitable candidate materials for use as a drug carrier via in situ gel formation and perform better than pluronics [9] due to their better biodegradability and structural integrity.

Besides diblock copolymers, also EO–CL–EO and CL–EO–CL triblocks have been described, and in some papers also stars containing different arms consisting of EO or CL [10].

In the literature there are numerous papers on the polymerization of lactones, in which different conditions for the polymerization are described. The polymerization has been performed using different catalysts [11], such as HCl in diethyl ether, alkoxylates of tin, lanthanum, calcium and other metals, sometimes also with metal complexes [12]. CL was also polymerized in presence of carboxylic acids without a metal catalyst [13] in order to avoid the presence of potentially toxic metal ions in the product.

Very often tin octoate is applied, as it has been approved by the Food and Drug Administration (FDA) [12].

The polymerization has also been carried out with microwave heating, which offers a number of advantages over conventional heating such as non contact heating, instantaneous and rapid heating and highly specific heating [14]. There has been some discussion about possible microwave effects [15–17].

Depending on the catalyst used the polymerization may proceed through different mechanisms (cationic, anionic, or coordinative), which may lead to products containing different amounts of unwanted molecules, such as linear or cyclic homopolymers etc.

Cyclic molecules are formed by a backbiting reaction, which can also occur without any catalyst, when PCL is heated over a longer period: the terminal hydroxy group may react with an ester group within the same chain, which leads to cyclic PCL oligomers, shorter block copolymers, and (in the extreme case) to the PEG homopolymer. Of course, there may also be intermolecular transesterification reactions.

According to the different nature of the active chain end, different side reactions may occur; consequently products with a different purity may be obtained.

As most of the applications mentioned above demand a high degree of purity, it appeared to be necessary to study the influence of reaction conditions (catalyst, temperature etc.) on the content of possible side products and residual starting materials. This required the development of reliable characterization methods, which yield the desired information.

2. Characterization of block copolymers

In the literature, the products are typically characterized using SEC, NMR, sometimes also by MALDI-TOF-MS. These techniques yield different information.

Size exclusion chromatography (SEC) can be used for determination of molar mass and molar mass distribution. If the product contains residual monomer or lower oligomers of CL, which are smaller than the initiator (PEG-MME), they can be determined by SEC. No information is,

however, obtained, whether the polymer peak contains the initiator or homopolymers besides the desired block copolymer.

NMR shows the presence of particular structural units (such as ether and ester groups) in the sample, it yields, however, no information on whether these groups are in the same or in different molecules. In other words, it yields just an average composition.

MALDI-TOF-MS is a powerful technique to characterize polymers with respect to their repeat unit and functionality, it is, however, not quantitative. Moreover, it is problematic in the characterization of samples with large polydispersities.

The required information – which includes the distributions of molar mass (MMD) and chemical composition (CCD) – can only be obtained by a combination of different techniques, which yield different kind of information.

Besides SEC, other chromatographic techniques may be applied: liquid adsorption chromatography (LAC) is mainly useful in the analysis of homopolymers. With block copolymers, where all blocks are polydisperse, overlapping series of peaks will be obtained.

Such samples can, however, be analyzed using a special mode of liquid chromatography, which allows the analysis of polymers according to structural units different from the repeat unit: liquid chromatography at critical conditions (LCCC)[18,19].

At a special mobile phase composition and temperature (the critical adsorption point, CAP) the retention volume of a polymer becomes independent of molar mass, which means, that a chain consisting of this particular repeat unit becomes chromatographically invisible [20,21].

This offers the opportunity to separate polydisperse polymers with respect to other structural units, such as end groups, other blocks etc. [22–25].

At the CAP for one repeat unit the corresponding (non-functional) homopolymer will elute at (or close to) the void volume. A polymer containing special end groups or another block can then be separated by different mechanisms. In principle, at the CAP for the main chain there may be different situations [21,26].

If the end groups or the other block of a copolymer interact with the surface of the stationary phase, all chains containing these structural units will elute in LAC mode (i.e. later than the homopolymer) [27,28]. This situation is useful in the separation of homopolymers according to their functionality. In the case of block copolymers, however, where the other block is polydisperse, the interaction with the stationary phase may be too strong, and thus the retention of higher oligomers may be too high.

If the other block is excluded from the pores, all chains containing this block will elute in SEC mode, i.e. earlier than the non-functional homopolymer. In the case of block copolymers (where the individual blocks are typically polydisperse), this situation is more favourable. It allows the determination of the length of the (non-critical) block and shows the presence of the critical homopolymer.

In the second case (one block at the CAP, the other one in SEC mode) one may separate the critical homopolymer from the copolymer (which coelutes with the excluded homopolymer)[29,30].

If critical conditions can be realized for both structural units (on different columns in different mobile phases), one may identify the homopolymers eluting at the CAP and determine the length of the excluded block.

This could be achieved for EO–PO block copolymers, as has been shown in a previous paper [26]: at the CAP for PO on a reversed phase column EO elutes in SEC mode, while at the CAP for EO on a normal phase column PO elutes in SEC mode. By combination of these techniques, the individual block length can be determined independently.

The situation is quite similar for EO–CL block copolymers: at the CAP for EO on the NP column one can also expect CL to elute in SEC mode. If a critical point for CL could be found on a reversed phase column, EO should also elute in SEC mode.

3. Experimental

3.1. Materials

Polyethylene glycol monomethyl ethers with an average molar mass of 2000 and 5000 (PEG–MME 2000 and PEG–MME 5000) and ϵ -caprolactone were purchased from Fluka (Buchs, Switzerland), tin(II) octoate, boron trifluoride ethyl etherate ($\text{BF}_3 \cdot \text{Et}_2\text{O}$), and NaH from Sigma–Aldrich (Vienna, Austria). The solvents used in chromatography were trichloromethane, acetone, THF, and water (all HPLC grade, from Carl Roth GmbH, Karlsruhe, Germany).

3.2. Polymerization

ϵ -Caprolactone was polymerized using $\text{BF}_3 \cdot \text{Et}_2\text{O}$, sodium hydride, and tin(II) octoate as catalysts and methanol, triethylene glycol monomethyl ether, PEG–MME 2000 and 5000 as initiators. All polymerizations were performed under Argon atmosphere.

Cationic polymerization (with $\text{BF}_3 \cdot \text{Et}_2\text{O}$) was performed in dichloromethane solution at room temperature (over night).

Anionic polymerization with sodium hydride was performed in THF solution. In this case the polymerization proceeded very rapidly: typically complete conversion was reached after a few minutes. In order to study the effect of heating on the formation of side products (by transesterification or backbiting), the reaction mixtures were also heated in microwave for up to 60 min at 140 °C. Anionic polymerizations were terminated by addition of acetic acid.

The other polymerizations were performed in bulk: the catalyst was added to the initiator (in the case of PEG–MME in melt), then the monomer was added. With tin octoate the polymerization was performed with microwave heating at 140 °C for 30 min in an Emrys ‘Optimizer’ Microwave instrument.

3.3. Spectroscopy

MALDI–TOF–Mass Spectrometry was performed on a Micromass TofSpec 2E Time-of-Flight Mass Spectrometer. The instrument is equipped with a nitrogen laser (337 nm

wavelength, operated at a frequency of 5 Hz), and a time lag focusing unit. Ions were generated by irradiation slightly above the threshold laser power. Positive ion spectra were recorded in reflectron mode applying an accelerating voltage of 20 kV and externally calibrated with a suitable mixture of poly(ethyleneglycol)s (PEG). The spectra of 100–150 shots were averaged to improve the signal-to-noise ratio. Analysis of data was done with MassLynx–Software V3.5 (Micromass/Waters, Manchester, UK).

Sample solutions were prepared by mixing a solution of dithranol ($c = 10 \text{ mg/mL}$ in THF), a solution of the analyte ($c = 2 \text{ mg/mL}$ in THF), and a solution of CF_3COONa (NaTFA; $c = 0.1 \text{ mg/mL}$ in THF), in a ratio of 7:1:1 (v/v/v). 0.5 μL of the resulting mixture were deposited on the sample plate (stainless steel) and allowed to dry under air.

3.4. Chromatography

Liquid chromatography was performed using the density detection system DDS 70 (CHROMTECH, Graz, Austria) which has been developed by our group. Data acquisition and processing was carried out by using software package CHROMA.

The solvents (chloroform, THF, acetone, methanol and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany) and Merck (Darmstadt, Germany).

Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria).

SEC measurements were performed on a modular SEC system comprising of a Gynkotek 300C pump (Gynkotek, Germering, Germany) equipped with a VICI injector (sample loop 100 μL) (from VICI, Baton Rouge, LA, USA), two column selection valves Rheodyne 7060 (Rheodyne, Cotati, CA, USA), a density detection system DDS 70 (Chromtech, Graz, Austria) coupled with an ERC 7512 RI detector (ERMA, Japan). SEC measurements were performed in chloroform on a PLgel 100 Å column (5 μm , 600 $\times$ 7.8 mm, Polymer Laboratories, Church Stretton, Shropshire, UK), and a Styragel HR3 column (10³ Å, 5 μ 300 $\times$ 7.8 mm, Waters, Milford, MA, USA). The flow rate in all SEC measurements was 1.00 ml/min, the column temperature 30.0 °C. Sample concentrations were 3.0–10.0 g/l.

LCCC measurements were performed at a temperature of 25.0 °C. The mobile phase was delivered by a JASCO 880 PU pump (Japan spectroscopic company, Tokyo, Japan) at a flow rate of 0.5 ml/min. Samples were injected manually using a Rheodyne 7125 injection valve equipped with a 50 μL loop. An ELSD (SEDEX 45, SEDERE, Alfortville, France) was coupled to the density detector. The evaporator temperature was 30 °C and pressure of carrier gas (nitrogen) was set to 1.0 bar.

The following columns were used in LCCC:

Jordi Gel DVB 500 RP: 100% poly(divinylbenzene); 250 $\times$ 4.6 mm; particle diameter = 5 μm ; nominal pore size = 500 Å, (Jordi, Bellingham, MA, USA).

Nucleosil 100-5 OH 5 μm , silica-based diol phase; 250 $\times$ 4.6 mm; particle diameter: 5 μm ; nominal pore size = 100 Å (Macherey–Nagel, Dueren, Germany).

Preparative chromatography was performed using the density detection system DDS 70 (CHROMTECH, Graz, Austria) which has been developed by our group. Data acquisition and processing was carried out by using software package CHROMA. The mobile phase was delivered by JASCO 880 PU pump (Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 2.0 ml/min. Samples were injected manually using Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a 500 μ l loop. A semi-preparative column was used for all measurements: Spherisorb ODS2 (25 $\times$ 300 mm, 5 μ m, 80 Å) from PhaseSep (Deeside, UK). The separations by semipreparative LCCC were performed at a temperature of 25 $^{\circ}$ C. Fractions were collected using an Advantec SF 2120 fraction collector (Advantec, Dublin, CA, USA). The column and density cell was placed in a thermostatted box, in which a temperature of 25.0 $^{\circ}$ C was maintained for all measurements using a thermostat Lauda RM6, Lauda-Königshofen, Germany). Mobile phases for LCCC were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602M (A. Paar, Graz, Austria).

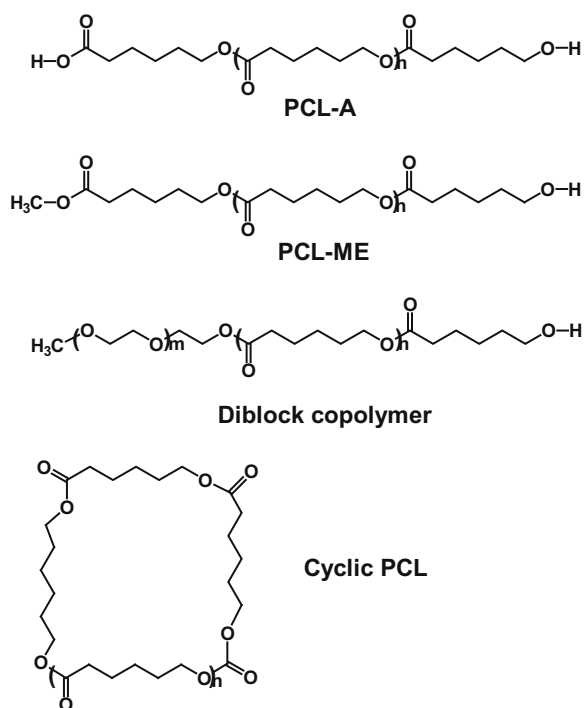
4. Results and discussion

In this study, we have synthesized diblock copolymers by polymerization of CL with PEG-MME as initiator. For comparison, homopolymers were synthesized with methanol or water as initiator. The resulting polymers have one hydroxy group at one end, while the other one is either a methyl ester or a carboxy group. In the following text we shall refer to these polymers as PCL methyl esters (PCL-ME) and PCL acid (PCL-A), as is shown in Scheme 1.

The first and rather easy question concerns the content of residual monomer. In principle, this can be achieved by SEC, which is commonly used to determine the molar mass distribution (MMD) of the copolymer.

Fig. 1 shows a comparison of the chromatograms of three EO-CL block copolymers (with different monomer-to-initiator ratio), the initiator and the monomer, which were obtained on a Styragel HR3 column (10³ Å, 5 μ 300 $\times$ 7.8 mm) in chloroform at a flow rate of 1.0 mL min⁻¹ with RI detection. On a column like this, which has sufficiently wide pores for the determination of the MMD, the monomer peak overlaps with the solvent peak, hence it cannot be determined quantitatively. The solvent peak cannot be avoided completely, as even presumably pure solvents typically contain small amounts of a stabilizer, moisture or other impurities, which lead to a vacancy peak resulting from preferential solvation of the polymer [31]. The content of residual monomer can, however, easily be determined by SEC on a column with rather narrow pores, on which, however, the copolymer and even the initiator (PEG-MME) may be beyond the exclusion limit.

Fig. 2 shows a comparison of the chromatograms of an EO-CL block copolymer, the initiator and the monomer, which were obtained on a PLgel 100 Å column (5 μ 600 $\times$ 7.8 mm) in chloroform at a flow rate of 1.0 mL min⁻¹ with RI detection. The peaks of the copolymer and the



Scheme 1. Possible structures in EO-CL diblock copolymers.

initiator almost coincide, as they are very close to the exclusion limit of the column. The monomer peak is, however, clearly separated from the solvent peak. As can be seen, there is still a considerable amount of residual monomer in this sample.

From the chromatograms shown in Fig. 1 one may calculate the MMD of the polymer. The increase in molar mass with the monomer-to-initiator ratio is obvious, but this does not necessarily mean, that the main peak is really the block copolymer: it may also contain residual initiator or the homopolymer of caprolactone.

SEC with dual detection (which works well for EO-PO copolymers [26,32,33]) cannot be applied in this case, as

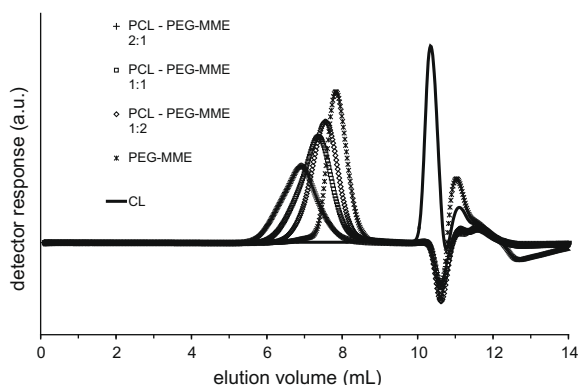


Fig. 1. SEC of 3 EO-CL block copolymers, the initiator (PEG-MME 2000) and the monomer, as obtained on the Styragel HR3 column (10³ Å, 5 μ 300 $\times$ 7.8 mm) in chloroform, detection: RI.

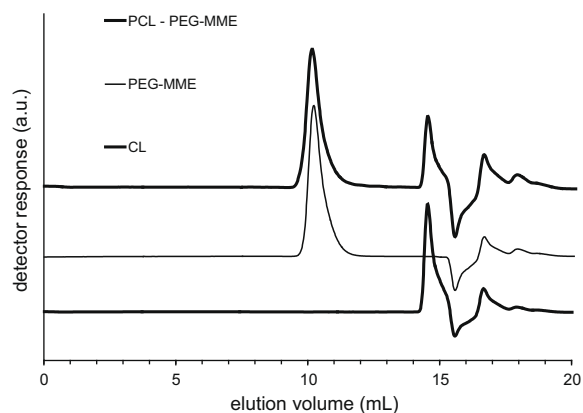


Fig. 2. SEC of an EO–CL diblock copolymer (1:1, as above), the initiator (PEG-MME 2000) and the monomer, as obtained on the PLgel 100 Å column (5 μ 600 $\times$ 7.8 mm) in chloroform, detection: RI.

the response factors of both repeat units in density and RI detection in the applicable solvents are too similar for a reliable determination of chemical composition. Anyway, this technique would only allow to identify a fraction as a homopolymer, if it contains 100% EO or CL.

The desired information can, however, be provided by LCCC. At the CAP for PEG on a normal phase column PPG elutes in SEC mode [26], and so does PCL. As can be seen in Fig. 3, PEG-monomethyl ethers elute on the Nucleosil 100-5 OH column in 77.01 wt-% acetone at the CAP, while PCL elutes in SEC mode. Under these conditions, CL–EO diblock copolymers with PEG-MME 2000 as initiator and different monomer-to-initiator ratios elute in SEC mode, while different PEG-MMEs elute at the same position, regardless their molar mass (Fig. 4).

In the next step we have tried to find critical conditions for PCL on a reversed phase column. As can be seen in Fig. 5, this could be realized on the Jordi Gel DVB 500 RP column in THF water (with 80.98 wt-% THF). In such a mobile phase, PCL-MEs elute at 2.8 mL, regardless their degree of polymerization. The peaks of PEG-MME and PCL-ME are well separated from each other. Obviously, PEG elutes in SEC mode (as PEG-MME 5000 elutes earlier than

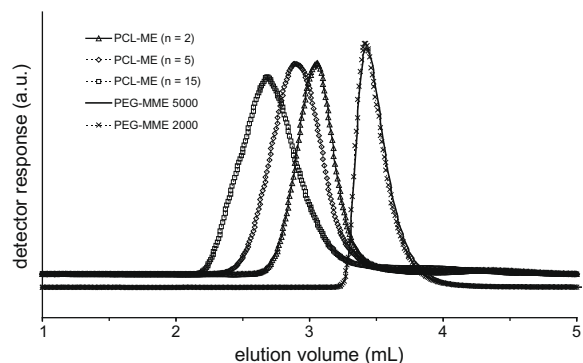


Fig. 3. LCCC of PEG-MME (at the CAP for EO) and PCL-ME with different degree of polymerization (prepared with tin octoate), as obtained on the Nucleosil 100-5 OH column in 77.01% acetone.

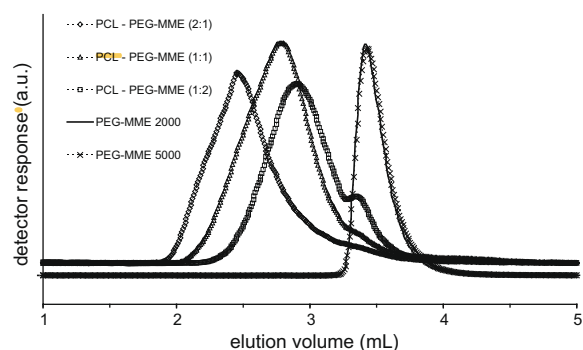


Fig. 4. LCCC of EO–CL diblock copolymers (prepared with tin octoate as catalyst and PEG-MME 2000 as initiator) and different PEG-monomethyl ethers at the CAP for EO (chromatographic conditions as above).

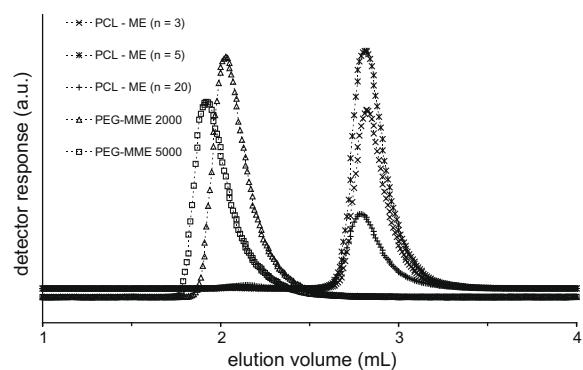


Fig. 5. LCCC of PEG-MME (at the CAP for CL) and PCL-ME (prepared with tin octoate as catalyst) with different degree of polymerization, as obtained on the Jordi column in 80.98% THF (ELS detection).

PEG-MME 2000). It must be mentioned, that PCL-ME at the CAP elutes later than the solvent peak (2.3 mL).

Fig. 6 shows the density trace of the same chromatograms. The monomer peak (which is invisible in ELS, but can be detected by the density detector) appears even later (at 3.1 mL) and is reasonably separated from the homopolymer PCL-ME.

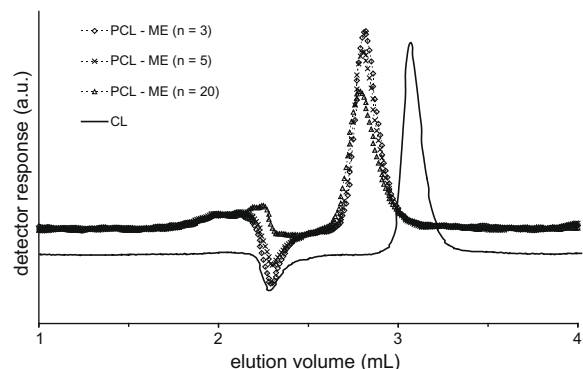


Fig. 6. LCCC (at the CAP for CL) of PCL-ME with different degree of polymerization (as in Fig. 5), as obtained on the Jordi column in 80.98% THF (density detection).

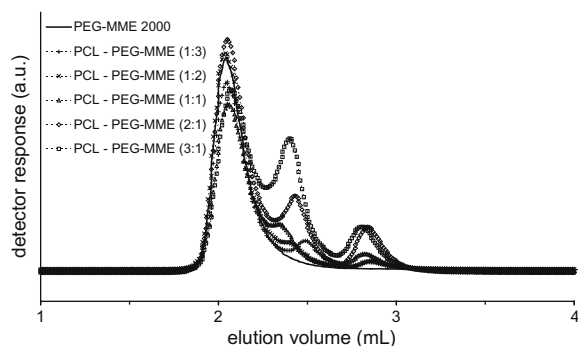


Fig. 7. LCCC of EO-CL block copolymers with different degree of polymerization (synthesized with $\text{BF}_3 \cdot \text{Et}_2\text{O}$ as catalyst and PEG-MME 2000 as initiator), as obtained on the Jordi column in 80.92% THF (ELS detection).

In block copolymers synthesized by polymerization of CL with PEG-MME as initiator and BF_3 -ethyl etherate as catalyst, the block copolymer elutes just like the initiator (as it should do at the CAP), but there are two additional peaks, which elute later: the second one at the position of the (linear) PCL-ME, and the first one between these two peaks.

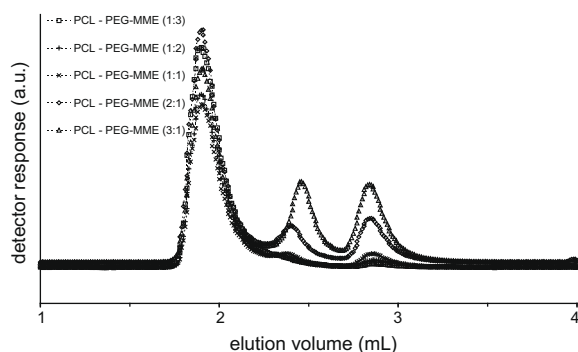


Fig. 8. LCCC of EO-CL block copolymers with different degree of polymerization (synthesized with $\text{BF}_3 \cdot \text{Et}_2\text{O}$ as catalyst and PEG-MME 5000 as initiator), as obtained on the Jordi column in 80.920% THF (ELS detection).

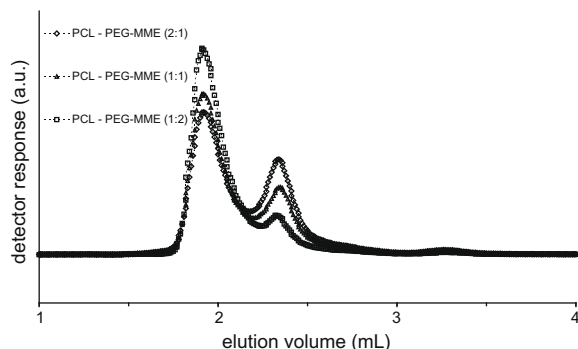


Fig. 9. LCCC of EO-CL block copolymers with different monomer-to-initiator ratio (synthesized with NaH as catalyst and PEG-MME 5000 as initiator), as obtained on the Jordi column in 80.98% THF (ELSD).

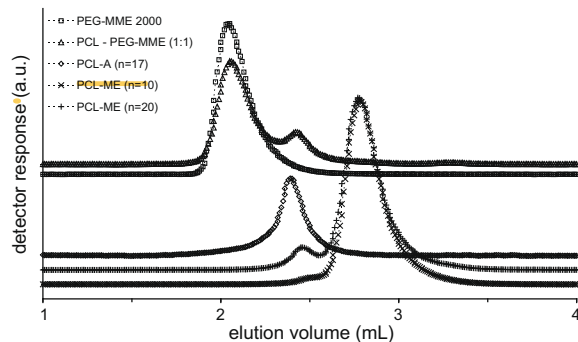


Fig. 10. LCCC (at the CAP for CL) of an EO-CL block copolymer (synthesized with NaH as catalyst and PEG-MME 2000 as initiator) and three homopolymers with different functionality, as obtained on the Jordi column in 80.98% THF (ELSD).

In the case of PEG-MME 2000 as initiator (Fig. 7) this fraction appears as a shoulder on the copolymer peak, in the samples obtained with PEG-MME 5000 (Fig. 8) as a well resolved peak at the same position. The amount of this fraction increases with the monomer-to-initiator ratio. One may speculate, that the second peak (which elutes roughly at the void volume) could be cyclic oligomers of CL, linear PCL with a carboxy group (PCL-A) from the reaction with water, or it may originate from degradation of the PEG chain (by backbiting).

The same behaviour is also observed in polymers obtained with sodium hydride as catalyst (Fig. 9). In this case, the third peak does not appear, but the second one is still present.

In a block copolymer initiated with PEG-MME 2000/sodium hydride the same pattern is observed. A homopolymer of PCL initiated with methanol-NaH and a monomer-to-initiator ratio of 20 also contained the fraction eluting at 2.4 mL (Fig. 10).

In anionic polymerization there should be no backbiting to the PEG chain, which could lead to shorter PEG chains. As the same peak appears also in products, which were obtained with methanol as initiator, this peak should rather contain cyclic oligomers of CL or linear chains with a different end group (most probably the carboxy group).

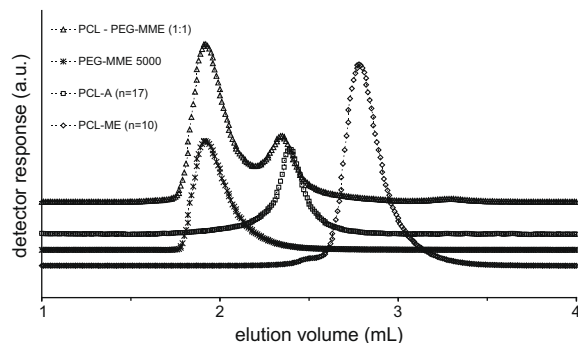


Fig. 11. LCCC (at the CAP for CL) of an EO-CL block copolymer (synthesized with NaH as catalyst and PEG-MME 5000 as initiator) and two homopolymers with different functionality, as obtained on the Jordi column in 80.98% THF (ELSD).

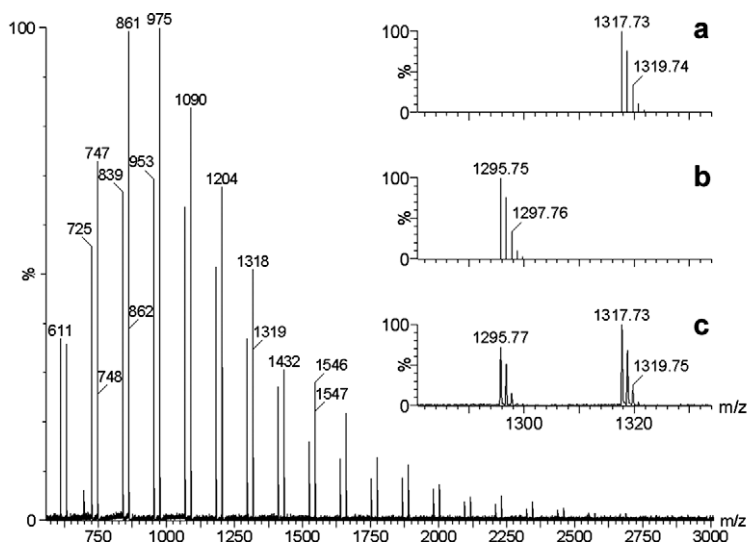


Fig. 12. MALDI-TOF-MS of a PCL-A showing two series of ions formed via H^+/Na^+ exchange at the carboxylic end group. Inset: comparison of the theoretical isotope pattern of $[(PCL_{(n=11)}-H + Na)^+ (C_{66}H_{111}O_{23}Na_2)]$ (a) and $[(PCL_{(n=11)}-A + Na)^+ (C_{66}H_{112}O_{23}Na)]$ (b), respectively, and the experimental result (c).

We have thus synthesized PCL-A by polymerization of CL with NaH and varying amounts of water. As can be seen in Fig. 11, the peak of this polymer appears at the same elution volume. This is a strong hint, that this peak is indeed PCL-A (formed by chain transfer to water). Of course there may also be cycles in this fraction. The proof can be given by MALDI-TOF-MS.

Fig. 12 shows the MALDI spectrum of a sample initiated with water/NaH. Evidently, there is just one homologous series, which can be identified as PCL-A.

Fig. 13 shows the spectrum of a sample obtained with methanol. As can be seen, this sample consists mainly of PCL-ME. Additionally, cyclic oligomers of PCL were ob-

served with low intensity in the mass range 500–1250 Da.

The situation is, however, more complex with EO-CL block copolymer samples, where both blocks are polydisperse, and where residual PEG-MME or side products can disturb the ionisation of the block copolymer significantly. As an example, a typical MALDI spectrum of such a crude product is reproduced in Fig. 14.

In principle, various expected signals of the EO-CL block copolymer were observed, but the spectrum is dominated by signals of cyclic PCL and residual PEG-MME due to strong mass discrimination. Consequently, a chromatographic fractionation prior to MALDI appeared necessary

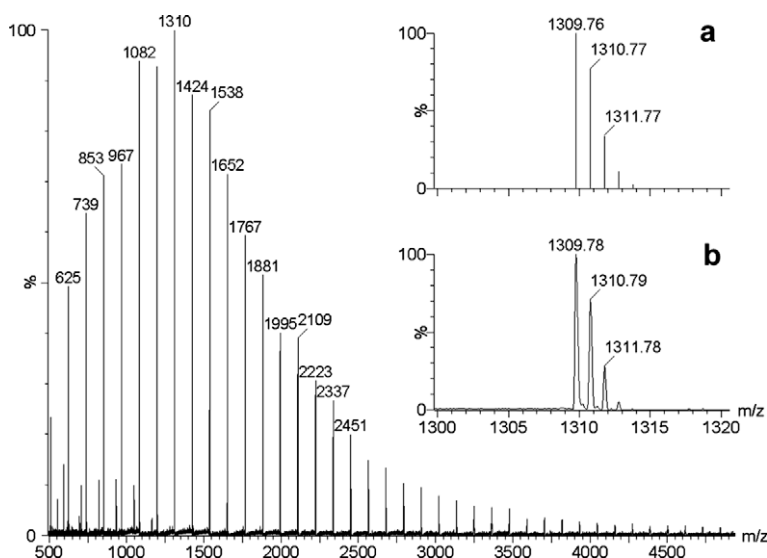


Fig. 13. MALDI-TOF-MS of a PCL-ME. Inset: comparison of the theoretical isotope pattern of $[PCL_{(n=11)}-ME + Na]^+ (C_{67}H_{114}O_{23}Na)$ (a) and the experimental result (b).

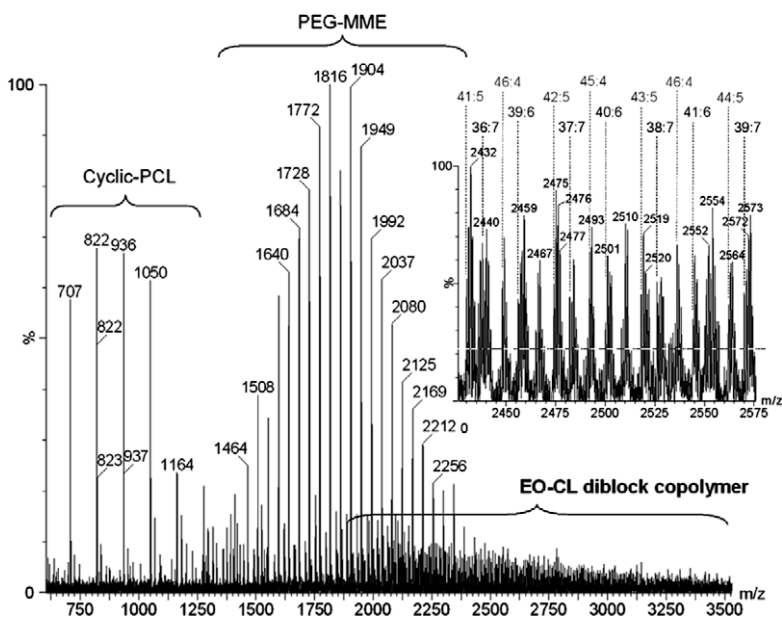


Fig. 14. MALDI-TOF-MS of a crude EO-CL diblock copolymer sample. Residual cyclic-PCL and PEG-MME 2000 dominate the spectrum due to mass discrimination. Inset: expanded view of the mass range 2425–2575 Da showing typical EO-CL signals. The labels $n:m$ indicate the monoisotopic mass of EO-CL macromolecules, where n is the degree of polymerization in the EO block, and m in the CL block, respectively.

for more detailed characterization of the EO-CL block copolymers.

As it is not possible to collect fractions behind the ELSD, semipreparative separation was performed on a different system. A Spherisorb ODS2 column was used at the CAP for CL, which was found to be in THF–water containing 80.48 wt-% THF as mobile phase.

The result of such a fractionation is shown in Fig. 15. Of course the resolution is not as good as on the analytical column, but the block copolymer is clearly separated from the homopolymers.

The MALDI spectrum of fraction 1 is shown in Fig. 16. As can be seen, this fraction contains the free acid (PCL-A) as also cyclic PCL. Fractions 2 and 3 are more or less the same (not shown).

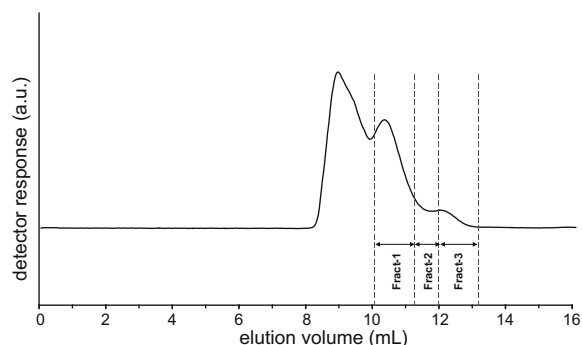


Fig. 15. Semipreparative LCCC of the EO-CL block copolymer shown in Fig. 11, as obtained on Spherisorb ODS2 (prep.) in THF–water with 80.48 Wt-% THF. Detection: density.

Evidently, the polymerizations with sodium hydride lead to considerable amounts of side products. Much cleaner block copolymers can be obtained with tin octoate as catalyst, as can be seen in Fig. 17. The MMD of such copolymers is typically unimodal.

While the polymerization with sodium hydride proceeds very rapidly, the reaction proceeds more slowly with tin octoate, it is complete after 30 min at 140 °C (in the microwave).

As the initiator must be molten before addition of the monomer and the catalyst, the temperature is still high

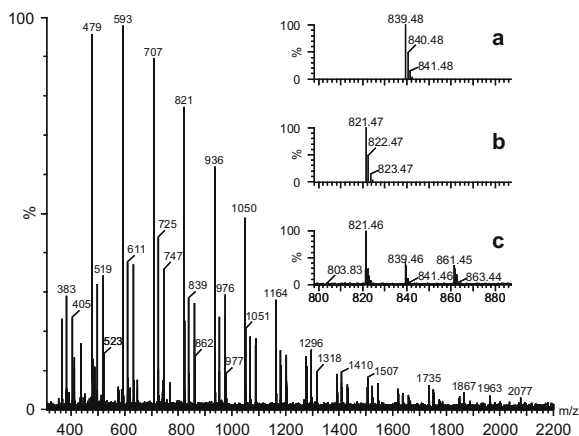


Fig. 16. MALDI-TOF-MS of fraction 1 from semipreparative LCCC (Fig. 12) showing cyclic PCL and PCL-A. Inset: comparison of the theoretical isotope pattern of $[(PCL_{(n=7)}A + Na)]^+$ ($C_{42}H_{72}O_{15}Na$) (a) $[cyclic-PCL_{(n=7)} + Na]^+$ ($C_{42}H_{70}O_{14}Na$) (b), and the experimental result (c). The third series of ions (e.g. at 861.45 Da) is sodiated PCL-A sodium salt.

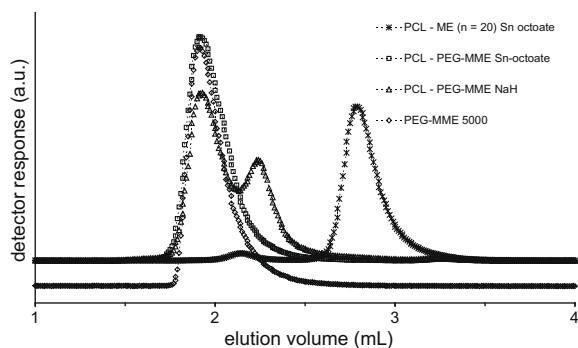


Fig. 17. LCCC of two EO–CL block copolymers (prepared with different catalysts) and the homopolymers, as obtained on the Jordi column in 80.98% THF (ELS detection).

enough (about 70 °C) for a rapid polymerization with NaH: it is often almost complete after a few minutes. Nevertheless we have heated the reaction mixture in microwave to 140 °C for 60 min to make sure, that all monomer is consumed. This should also show, whether transesterification reactions will take place under such conditions. As can be seen in Fig. 18, the shoulder is quite pronounced after only 10 min, increases till 30 min, and remains almost constant

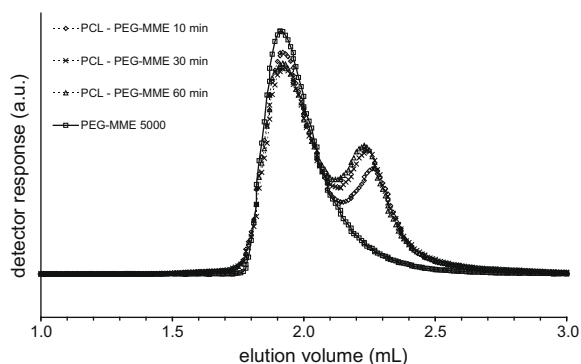


Fig. 18. LCCC of EO–CL block copolymers (prepared with NaH as catalyst and PEG-MME 5000 as initiator) at different reaction times, as obtained on the Jordi column in 80.98% THF (ELS detection).

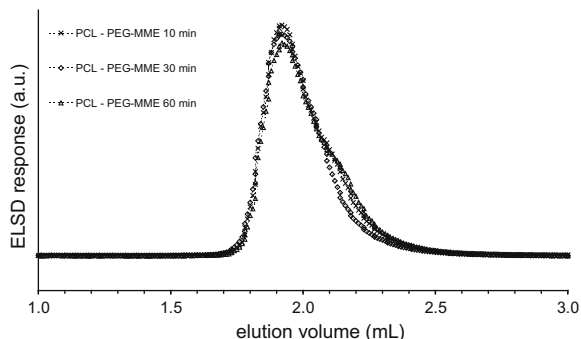


Fig. 19. LCCC of EO–CL block copolymers (prepared with tin octoate) at different reaction times, as obtained on the Jordi column in 80.98% THF (ELS detection).

until 60 min. With tin octoate, the peak of PCL-A does not appear even after longer heating (Fig. 19).

5. Conclusions

Block copolymers of ethylene oxide and ϵ -caprolactone can be characterized using size exclusion chromatography (SEC) and liquid chromatography at critical conditions (LCCC). SEC yields the overall MMD of the products, the even more important information on chemical heterogeneity can be provided by LCCC. On a normal phase column, critical conditions for PEO can be realized. In such a mobile phase PCL elutes in SEC mode, which allows a determination of the length of the PCL block and the identification of unreacted PEG (if present in the polymer). On a reversed phase column, critical conditions for CL have been found. Under such conditions, PEO elutes in SEC mode, which allows a determination of the length of the EO block, and the identification of CL homopolymers in the block copolymer.

Moreover, CL homopolymers with different functionality can also be separated from each other. The nature and amount of side products depends strongly on the catalyst used in the polymerization. The highest purity of the block copolymers is obtained with tin octoate.

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